

Solution

SOUND-03

Class 09 - Science

Section A

1. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: When the force retards the motion, the force and displacement are in opposite directions to each other. Hence, the work done by the force is negative.
2. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Since, power is inversely proportional to time, crane P supplies more power.
3. **(c)** A is true but R is false.
Explanation: For conservative forces the sum of kinetic and potential energies at any point remains constant throughout the motion. It does not depend upon time. This is known as law of conservation of mechanical energy. According to this rule.
Kinetic energy + Potential energy = E = constant
4. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Since power is inversely proportional to time, crane P supplies more power.
5. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The work done on the spring against the restoring force is stored as potential energy in both cases when it is compressed or stretched.
6. **(d)** A is false but R is true.
Explanation: Potential energy (u) = $\frac{1}{2} kx^2$
i.e. $u \propto x^2$
This is an equation of the parabola, so graph between u and x is a parabola, not a straight line.
7. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: From, definition, work done in moving a body against a conservative force is independent of the path followed.
8. **(c)** A is true but R is false.
Explanation: Kilowatt (1000 watt) is the unit of power. Kilowatt hour is the power consumed in one hour = power $\times$ time = energy. One watt is equal to Joule per second. Hence, watt $\times$ time has units of energy.
9. **(d)** A is false but R is true.
Explanation: The work done by a non-conservative force in a round trip is not zero. Since the body moves, there must be force acting on the body.
10. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
11. **(c)** A is true but R is false.
Explanation: Work done on the body in order to increase its velocity v_1 to v_2 is given by,

$$\begin{aligned} dW &= \int_{v_1}^{v_2} F ds = \int_{v_1}^{v_2} m \frac{dv}{dt} ds \\ &= \int_{v_1}^{v_2} m \left(\frac{ds}{dt} \right) dv = \int_{v_1}^{v_2} mv dv \\ &= m \left[\frac{v^2}{2} \right]_{v_1}^{v_2} = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 \end{aligned}$$

Thus, work done by a force acting on a body is equal to a change in kinetic energy of the body. This is true for a system containing any number of particles.

12. **(d)** A is false but R is true.
Explanation: Kinetic energy of a body of mass m_1 ,

$$K_1 = \frac{1}{2}mv_1^2 = \frac{p_1^2}{2m_1}$$
 Again, kinetic energy of a body of mass m_2 ,

$$K_2 = \frac{1}{2}mv_2^2 = \frac{p_2^2}{2m_2}$$
 If $p_1 = p_2$, $\frac{K_1}{K_2} = \frac{m_2}{m_1}$
 As given $m_2 > m_1$
 Therefore, $K_1 > K_2$ i.e. the kinetic energy of light body will be more than the kinetic energy of heavy body when both have same momentum.
13. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
14. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Work done $W = \vec{F} \cdot \vec{ds} = Fds \cos \theta = 0$ when $\theta = 90^\circ$
 No work is done when force is normal to the displacement.
15. **(c)** A is true but R is false.
Explanation: When a toy car is winded, work done on it gets stored in the form of elastic potential energy and when it is placed on the ground, this elastic potential energy slowly converts into kinetic energy of the toy and facilitates its motion.
16. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Work done by the gravitational force is mgh .

Section B

17. State True or False:
- (a)** True
Explanation: True
 - (a)** True
Explanation: True
 - (b)** False
Explanation: False. The energy possessed by a body because of its position or configuration is known as potential energy.
 - (a)** True
Explanation: True
 - (a)** True
Explanation: True
 - (a)** True
Explanation: True
 - (b)** False
Explanation: False. Work done is positive when the force is in the direction of displacement.
 - (a)** True
Explanation: True
 - (b)** False
Explanation: False. The commercial unit of energy is kilowatt hour.
 - (b)** False
Explanation: False. Moving water can do work by turning the blades of a turbine for generating electricity is an example of kinetic energy.

(xi) (a) True

Explanation: True

(xii) (a) True

Explanation: True

(xiii) (a) True

Explanation: True

(xiv) (a) True

Explanation: True

(xv) (b) False

Explanation: False. The SI unit of work is joule.

(xvi) (b) False

Explanation: False. The energy used in households, industries, and commercial establishments is usually expressed in kilowatt-hour.

(xvii) (a) True

Explanation: True

(xviii)(a) True

Explanation: True

Section C

18.

(c) $P_1 < P_2$

Explanation: Smaller the area of contact greater will be the pressure exerted as $P \propto \frac{1}{A}$

19.

(c) activity (ii) only

Explanation: More impression will be made by cuboid in activity (ii) as $P \propto \frac{1}{A}$

Section D

20.

(c) 1-D, 2-A, 3-C, 4-B

Explanation:

- i. Kinetic energy is the energy of motion.
- ii. Energy possessed by an object because of its position is called potential energy.
- iii. Work is defined as a force acting upon an object to cause a displacement .
- iv. Rate of doing work is called power.

21.

(c) 1-D, 2-A, 3-C, 4-B

Explanation:

- i. Kinetic energy is the energy of motion. $KE = \frac{1}{2} mv^2$
- ii. Energy possessed by an object because of its position is called potential energy.
- iii. Work is defined as a force acting upon an object to cause a displacement .
- iv. Rate of doing work is called power.

22.

(d) 1-B, 2-D, 3-A, 4-C

Explanation:

- i. Law of conservation of energy is stated as energy neither be created nor be destroyed but it converted into one form to another form.
- ii. Stretched rubber band possess potential energy.
- iii. At mean position pendulum attains maximum kinetic energy.
- iv. Various energy forms are Inter convertible.

23.

(b) 1-A, 2-C, 3-B, 4-D

Explanation:

- i. As the velocity of object increases its Kinetic energy also increases.
- ii. $W = F \times S \cos \theta$ ($\cos 0^\circ = 1$) So, $w = F \times S$ means maximum work is done.
- iii. $W = F \times S \cos \theta$ ($\cos 90^\circ = 0$) So, $w = 0$ means zero work is done.
- iv. $W = F \times S \cos \theta$ where F = force, d = displacement

24.

(c) 1-B, 2-D, 3-A, 4-C

Explanation:

- i. In Battery, chemical energy converts into electrical energy.
- ii. In Bulb, electrical energy converts into light energy.
- iii. In Steam engine, steam energy converts into mechanical energy.
- iv. In microphone, sound energy converts into electrical energy.

25.

(b) 1-B, 2-D, 3-A, 4-C

Explanation:

- i. Kinetic energy is the energy of motion. $KE = \frac{1}{2} mv^2$
- ii. Energy possessed by an object because of its position is called potential energy. $P.E. = mgh$
- iii. Rate of doing work is called power. $P = \frac{W}{t}$
- iv. Energy = Power $\times$ time, $e = P \times t$

26.

(d) 1-D, 2-A, 3-C, 4-B

Explanation:

- i. Bullet fired from gun is kinetic energy because Kinetic energy is the energy of motion.
- ii. A flying bird depends on motion and position also so its example of potential and kinetic energy.
- iii. In a falling stone potential energy changes to kinetic energy
- iv. Water at dam stored in particular position so, it's an example of potential energy.

27.

(b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

- Work is defined as a force acting upon an object to cause a displacement. The unit of work is Joule.
- The rate of doing work is called power. The SI unit of power is the watt.
- Energy is the capacity of doing work. A large quantity of energy is expressed in kilowatt-hour and is written as kWh.
- The unit of force is Newton.

28.

(b) 1-C, 2-B, 3-D, 4-A

Explanation:

- i. Work is defined as a force acting upon an object to cause a displacement. Unit of work is Joule.
- ii. Rate of doing work is called power. The SI unit of power is watt.
- iii. Energy is the capacity of doing work. Large quantity of energy is expressed in kilowatt hour and is written as kWh.
- iv. Unit of force is newton.

29.

(d) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

- Kinetic energy is the energy of motion. $KE = \frac{1}{2} mv^2$
- The energy possessed by an object because of its position is called potential energy. $P.E. = mgh$
- The rate of doing work is called power. $P = \frac{w}{t}$
- $Energy = Power \times time$, $E = P \times t$

30.

(c) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

Explanation:

- The SI unit of work is joule and is denoted as 'J', which is named after an English physicist James Prescott Joule. The 1 joule of work done is equal to $1N \times 1 m$. Or, 1 joule = Nm
- 1 kilowatt = 1000 watt
- $1 kWh = 1 kW \times 1h$
Or, $1kWh = 1000 \text{ joule} \times 3600 s$
Or, $1 kWh = 3600000 \text{ joule}$
Or, $1kWh = 3.6 \times 10^6 \text{ joule}$
- 1 HP = 746 W

31.

(d) 1-D, 2-A, 3-C, 4-B

Explanation: (1) Nm is also called Erg.

(2) In free fall potential energy is equal to kinetic energy

(3) A donkey is carrying a load on its back, the force of gravity on the load is in the downward direction, whereas the displacement is in the horizontal direction, i.e., the force and the displacement are perpendicular to each other. There is no displacement in the direction of the force of gravity, and therefore, work done is zero.

(4) An engine pulling a train on a railway line is said to be doing work.

32. **(a)** (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Explanation:

- Bullet fired from guns is kinetic energy because Kinetic energy is the energy of motion.
- A flying bird depends on motion and position also so its example of potential and kinetic energy.
- In a falling stone potential energy changes to kinetic energy.
- Water at dam stored in a particular position so, it's an example of potential energy.

33.

(c) 1-A, 2-C, 3-B, 4-D

Explanation: (1) The SI unit of work is joule and is denoted as 'J', which is named after an English physicist James Prescott Joule. The 1 joule of work done is equal to $1N \times 1 m$. Or, 1 joule = Nm

(2) 1 kilowatt = 1000 watt

(3) $1 kWh = 1 kW \times 1 h$

Or, $1kWh = 1000 \text{ joule} \times 3600 s$

Or, $1 kWh = 3600000 \text{ joule}$

Or, $1kWh = 3.6 \times 10^6 \text{ joule}$

(4) 1 HP = 746 W

34.

(d) 1-C, 2-B, 3-D, 4-A

Explanation: (1) Law of conservation of energy is stated as energy neither be created nor be destroyed but it converted into one form to another form.

- (2) Stretched rubber band possess potential energy .
 (3) At mean position pendulum attains maximum kinetic energy.
 (4) Various energy forms are inter convertible.

35.

(b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

- Nm is also called Erg.
- In free fall potential energy is equal to kinetic energy.
- A donkey is carrying a load on its back, the force of gravity on the load is in the downward direction, whereas the displacement is in the horizontal direction, i.e., the force and the displacement are perpendicular to each other. There is no displacement in the direction of the force of gravity, and therefore, work done is zero.
- An engine pulling a train on a railway line is said to be doing work.

36.

(c) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

- In Battery, chemical energy converts into electrical energy.
- In Bulb, electrical energy converts into light energy.
- In the Steam engine, steam energy converts into mechanical energy.
- In microphones, sound energy converts into electrical energy.

Section E

37. **(a)** statement B

Explanation: The average kinetic energy of a gas particle is directly proportional to the temperature. An increase in temperature increases the speed at which the gas molecules move. So statement A is a false statement.

As the body falls from height h , its potential energy decreases while the kinetic energy begins to increase; the total mechanical energy at any point being the same i.e. mgh . So, statement B is a true statement.

38.

(c) Statement B

Explanation: In general, work represents any physical or mental activity. But work has a precise meaning in physics. Work is said to be done when a net force acts on a body and the body undergoes displacement. So statement A is false.

Work done by a force is positive if the applied force has a component in the direction of the displacement. So, statement B is true.

39.

(d) statement B

Explanation:

A. Mechanical energy is not always conserved. Mechanical energy is the sum of kinetic and potential energy in a system.

Mechanical energy is conserved so long as we ignore air resistance, friction, etc. When we don't ignore outside forces, such as those just mentioned, mechanical energy is not conserved.

B. According to the law of conservation of energy the total energy of the system is always conserved.

So, Statement B is true statement and Statement A is a false statement.

40.

(c) neither statement A nor statement B is true.

Explanation:

A. Two bodies having equal kinetic energy may or may not have equal momentum.

B. The work done by a centripetal force is always zero.

So, statements A and B both are not true statements.

41.

(d) Statement B

Explanation: Work is said to be done when there is a displacement of the body with respect to moving body. No work is done when a camel is carrying a load on its back. But work is done when a sailboat is moving due to wind energy. So, statement B is a true statement.

42.

(d) statement B

Explanation: The aeroplane has both kinetic energy and potential energy. When it is in flight it will fly above the ground so it is displaced by certain distance and hence it possesses potential energy. As it is in motion it possesses kinetic energy.

Section F

43. i. (a) 4 times

ii. (c) Only (II)

iii. (a) $\frac{1}{2} mv$

iv. (c) 120J

v. (a) kinetic energy

44. i. (c) position and configuration

ii. (a) kinetic energy to heat energy

iii. (a) increase

iv. (d) potential energy

v. (c) half potential and half kinetic energy

45. i. (a) watt

ii. (c) both (a) and (b)

iii. (a) Only (I)

iv. (b) 0.36 unit

Given: $P = 60 \text{ W} = 0.06 \text{ kW}$; $t = 6 \text{ hrs}$

Thus energy consumed per day is:

$E = Pt = 0.06 \times 6 = 0.36 \text{ kWh} = 0.36 \text{ units}$

v. (a) III and IV

46. i. (d) Only (II)

ii. (a) 1000J

iii. (a) created, destroyed

iv. (a) conversion of K. E into heat energy.

v. (b) remain unchanged

47. i. (a) 10J

ii. (d) 180°

iii. (d) initial velocity of the object

iv. (b) (I) and (II)

v. (a) zero work done

48. i. **Potential Energy**- It is the energy possessed by a body by virtue of its position or shape.

ii. In a toy car, the wound spring possesses potential energy. When spring is released, its potential energy changes into kinetic energy due to which the toy car moves.

iii. Given, $m = 50 \text{ kg}$, $h_1 = 75 \text{ m}$, $h_2 = 60 \text{ m}$

At point A, $PE_1 = mgh_1 = 50 \times 10 \times 75$

$= 37500 \text{ J}$

At point B, $PE_2 = mgh_2 = 50 \times 10 \times 60$

$= 30000 \text{ J}$

Change in PE $= PE_1 - PE_2 = 37500 - 30000 = 7500 \text{ J}$

49. i. When it strikes ground, its PE is zero and after bouncing, its potential energy increases gradually.

ii. At the time it strikes the ground, it has maximum KE and after it bounces, its KE starts changing into potential energy.

iii. The ultimate or total energy remains constant at any point of time during the motion.

50. i. **Law of conservation of energy:** The energy in a system can neither be created nor destroyed. It may be transformed from one form to another, but the total energy of the system remains constant.
- ii. The energy possessed by a body due to a displacement caused in it by the application of a force is called mechanical energy.
- iii. Energy = Power $\times$ Time = $4 \times 600 \text{ W} \times 5 \text{ h} = \frac{4 \times 600 \times 5}{1000} = \frac{12000}{1000} = 12 \text{ KWh}$
- Hence the energy in kWh consumed is 12 kWh.
51. Given, mass of the vehicle, $m = 1 \text{ tonne} = 1000 \text{ kg}$
- Initial speed, $u = 60 \text{ ms}^{-1}$
- Distance between vehicle and the cow, $s = 9 \text{ m}$
- Final velocity, $v = 0$
- i. KE of vehicle before applying brakes is given by = $\frac{1}{2} mu^2 = \frac{1}{2} \times 1000 \times 60 \times 60 = 1800000 \text{ J} = 1800 \text{ kJ}$
- ii. From the third equation of motion,
- $$v^2 - u^2 = 2as$$
- $$(0)^2 - (60)^2 = 2 \times a \times 9$$
- $$\Rightarrow a = -200 \text{ ms}^{-2}$$
- So, retarding force provided by the brakes = $ma = 1000 \times (-200) = -200000 \text{ N}$
- iii. Now, again from the second equation of motion,
- $$s = ut + \frac{1}{2} at^2 \Rightarrow 9 = 60t + \frac{1}{2} \times (-200) t^2$$
- or $9 = 60t - 100t^2$
- or $100t^2 - 60t + 9 = 0 \Rightarrow (10t - 3)^2 = 0$
- or $10t - 3 = 0 \Rightarrow t = \frac{3}{10} = 0.3 \text{ s}$
- iv. So, work done by the braking force is given by = $Fs = -200000 \times 9 = -1800000 \text{ J} = -1800 \text{ kJ}$
52. Mass = $m = 5 \text{ Kg}$
- Height = $h = 12 \text{ m}$.
- g = Acceleration due to gravity = 9.8 m/s^2
- i. P.E. = mgh
- $$= 5 \times 12 \times 9.8$$
- $$= 588 \text{ J}$$
- ii. Force = ?
- Work Done = P. E. energy of the Body
- Force $\times$ Distance Moved = 588
- $$F \times 12 = 588$$
- $$F = 49 \text{ N}$$
- iii. The work done in lifting the body gets stored as the potential energy.
53. i. When a force of 1 Newton causes a displacement of 1 m in its own direction, the work done is said to be one joule.
- ii. $p = 5000 \text{ W}$, $t = 2 \text{ h}$
- Energy consumed = $pt = 5000 \times 2 = 10000 \text{ Wh} = 10 \text{ kW} = 10 \text{ units}$.
- iii. When a carpet is beaten with a stick, the dust comes out of it because of the law of inertia. Initially, the dust particles are at rest along with the carpet. Beating the carpet with the stick makes the carpet move but the dust particles remain at rest due to inertia at rest, thus the dust gets detached from the carpet.
54. Given, mass of the object, $m = 10 \text{ kg}$
- Height, $h = 50 \text{ cm} = 0.5 \text{ m}$
- i. As potential energy is given by $PE = mgh$
- $$= 10 \times 10 \times 0.5$$
- $$= 50 \text{ J}$$
- ii. From the law of conservation of energy, the total energy of the ball just before dropping = total energy of the ball just on touching the ground
- $$\Rightarrow KE + PE \text{ of the ball just before dropping}$$
- $$\Rightarrow KE = 50 \text{ J}$$
- iii. As we know, $KE = 50 \text{ J}$
- So, $\frac{1}{2} mv^2 = 50 \Rightarrow v^2 = \frac{50 \times 2}{10} = 10$

So, the velocity with which it hits the ground,

$$v = \sqrt{10} = 3.16 \text{ ms}^{-1}$$

55. (a) $F = 250 \text{ kg} \times 10 \text{ ms}^{-2}$ ($g = 10 \text{ ms}^{-2}$) $= 2500 \text{ N}$ $s = 1 \text{ m}$ $W = F.s = 2500 \text{ N} \times 1 \text{ m} = 2500 \text{ Nm} = 2500 \text{ J}$ (b) Zero, as the box does not move at all while holding it. (c) In order to hold the box, men are applying a force which is opposite and equal to the gravitational force acting on the box. While applying the force, muscular effort is involved. So, they feel tired.
56. i. The potential energy of the car remains the same since $PE (=mgh)$ is independent of velocity.
ii. The kinetic energy of the car becomes four times, since $KE = (\frac{1}{2}mv^2)$ is proportional to square of velocity.
iii. The momentum of the car will also get doubled since momentum ($p = mv$) is proportional to velocity.
57. i. Kinetic energy $= \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 900 \times (30)^2 = \frac{1}{2} \times 900 \times 900$
 $= 4,05,000 \text{ J}$
ii. Energy used = Work done against resistive force
 $= \text{Force} \times \text{Distance}$
 $= 2,000 \times 30 = 60,000 \text{ J} = 60 \text{ kJ}$
iii. Minimum power $= \frac{\text{Energy used}}{\text{Time taken}}$
 $= \frac{60,000 \text{ J}}{1 \text{ s}}$
 $60,000 \text{ W} = 60 \text{ W}$
iv. Chemical energy. The *fuel* in your *car* provides chemical potential *energy*, which is converted to kinetic *energy* when it's combusted.